SYSDESIGN REFERENCE

# Distributed ID Generation

Core concepts, reference diagram, tradeoffs, and interview prompts.

Create unique, scalable identifiers with acceptable ordering, coordination, and information leakage.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / ID-GENERATION

SYSDESIGN

# Distributed ID Generation

**CORE IDEA**

**Create unique, scalable identifiers with acceptable ordering, coordination, and information leakage.**

## Reference architecture decision chain

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| Uuid Variants | Database Sequences | Snowflake-Style Ids | Clock Behavior |

### Decisions to defend

+ UUID variants
+ database sequences
+ Snowflake-style IDs
+ clock behavior
+ and shard locality.

### Failure modes to test

+ Clock rollback
+ duplicate worker IDs
+ hotspotting
+ and guessable IDs.

## Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

# Distributed ID Generation

| Probe | What a strong answer makes explicit |
| --- | --- |
| How does the design behave during a regional dependency failure? | Requirement, decision, cost, failure behavior, and observable metric. |
| Which metric reveals the first bottleneck? | Requirement, decision, cost, failure behavior, and observable metric. |
| What consistency does the user actually observe? | Requirement, decision, cost, failure behavior, and observable metric. |
| How would the design change at ten times the traffic? | Requirement, decision, cost, failure behavior, and observable metric. |

## Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

## Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

**DESIGN DISCIPLINE**

**For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?**

PUBLISHING STATUS

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